

Semi-Supervised Latent Variable Model
for NYC Property Valuation

Motivation & Problem

Property valuation underpins **tax administration** to ensure fair revenue and **urban planning** to optimize resource allocation. We target both: taxation requires precise financial estimates, while planning uses value gradients to identify development zones.

Challenges:

- **Heavy-tailed Distribution:** Prices exhibit excess kurtosis of 9.27, violating standard Gaussian assumptions.
- **Missingness:** About 25% of records lack price data.

Goal: Learn a latent representation to predict sale prices and provide uncertainty estimates without discarding incomplete data.

Methodology

We implement a **Semi-Supervised MIWAE** (SemiSupMIWAE) using a Student- t Mixture Prior.

Generative Model

- **Latent Space:** $z_i \sim \sum_{k=1}^K \pi_k t_i(\mu_k, \Sigma_k)$.
- **Covariates:** $x_i \mid z_i \sim \mathcal{N}(\mu_x(z_i), \text{diag}(\sigma_x^2(z_i)))$.
- **Price Head:** $y_i \mid z_i \sim \mathcal{N}(\mu_y(z_i), \sigma_y^2(z_i))$.

Inference We maximize the importance-weighted lower bound, the ELBO:

$$\mathcal{L} \approx \mathbb{E}_{q_\phi} \left[\log \frac{1}{K_{IW}} \sum_{k=1}^{K_{IW}} \frac{p_\theta(x_i, y_i \mid z_i^{(k)}) p(z_i^{(k)})}{q_\phi(z_i^{(k)} \mid x_i, y_i)} \right]$$

Graphical Model

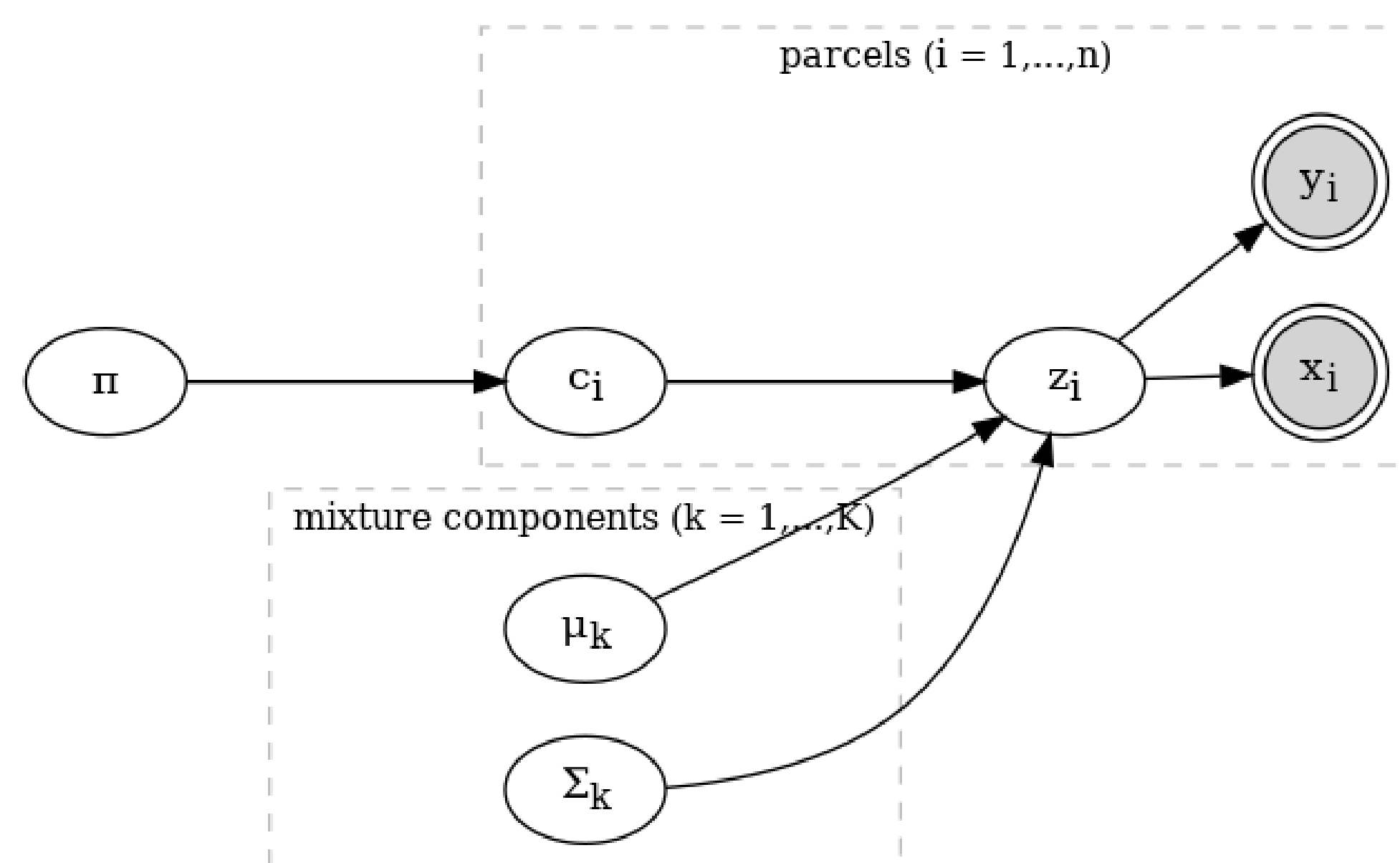


Figure: SemiSupMIWAE Structure.

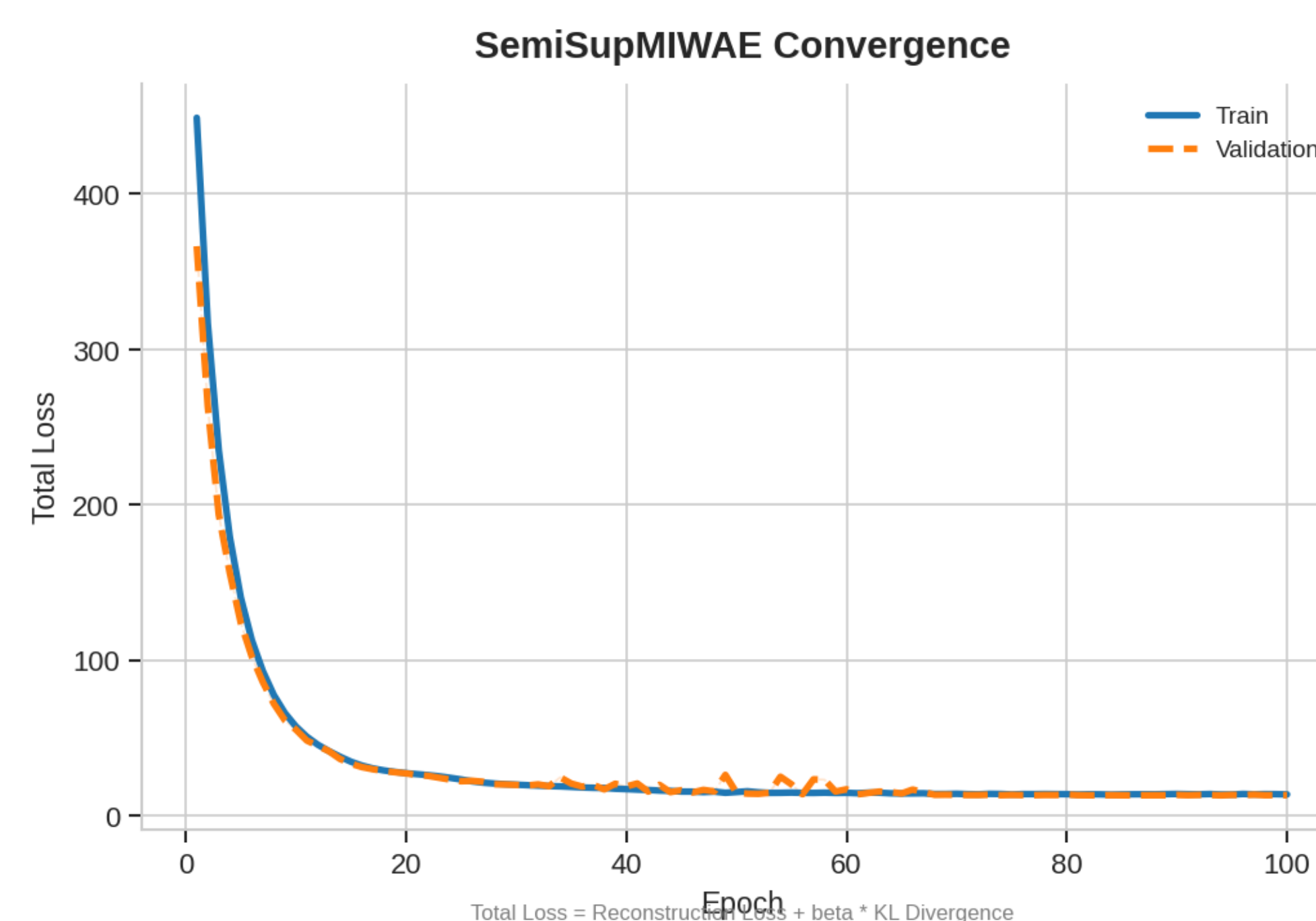


Figure: Convergence (Neg. ELBO) over 50 epochs.

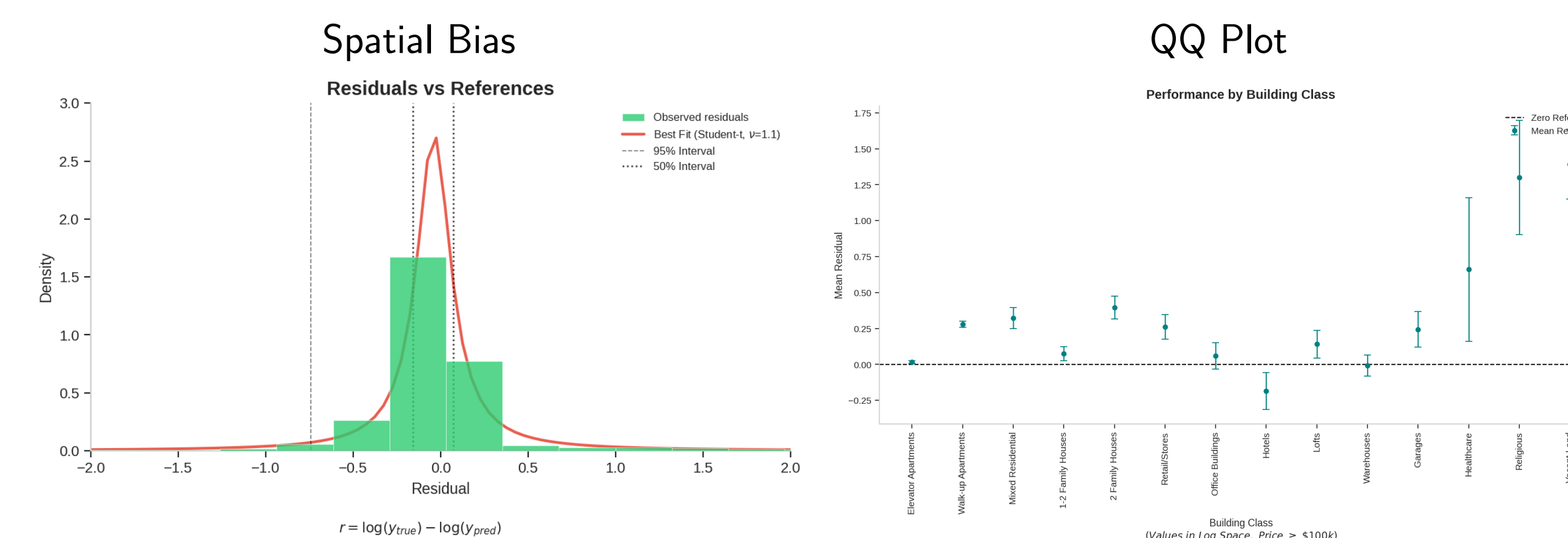
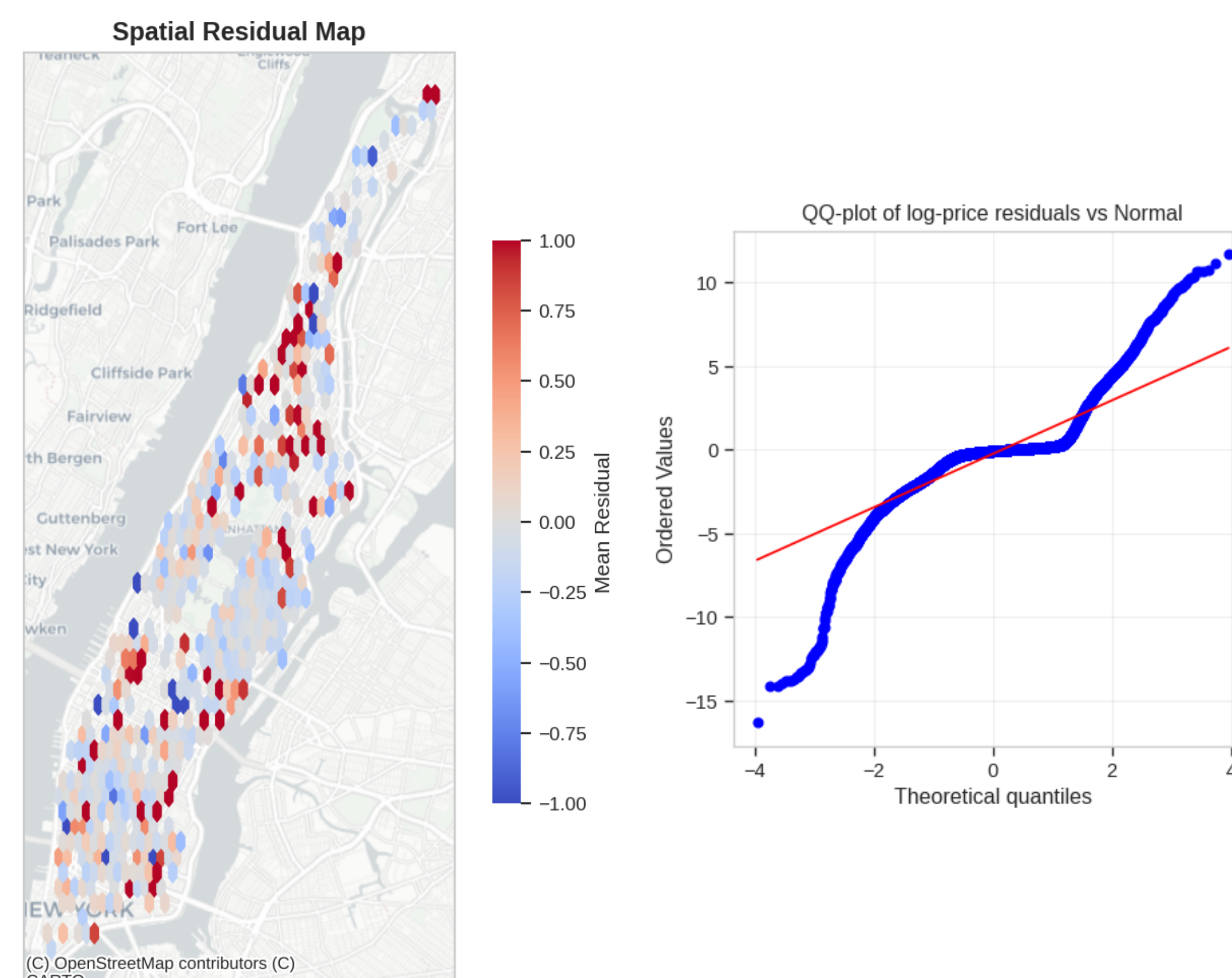
Predictive Performance

Evaluated on a 20% held-out set.

Metric	Value
Avg. Log Posterior $p(y \mid x)$	-2.28
RMSE	\$24.6 Million
MAE	\$2.18 Million
Median Absolute Error	\$128,000
95th Percentile Error	\$60 Million

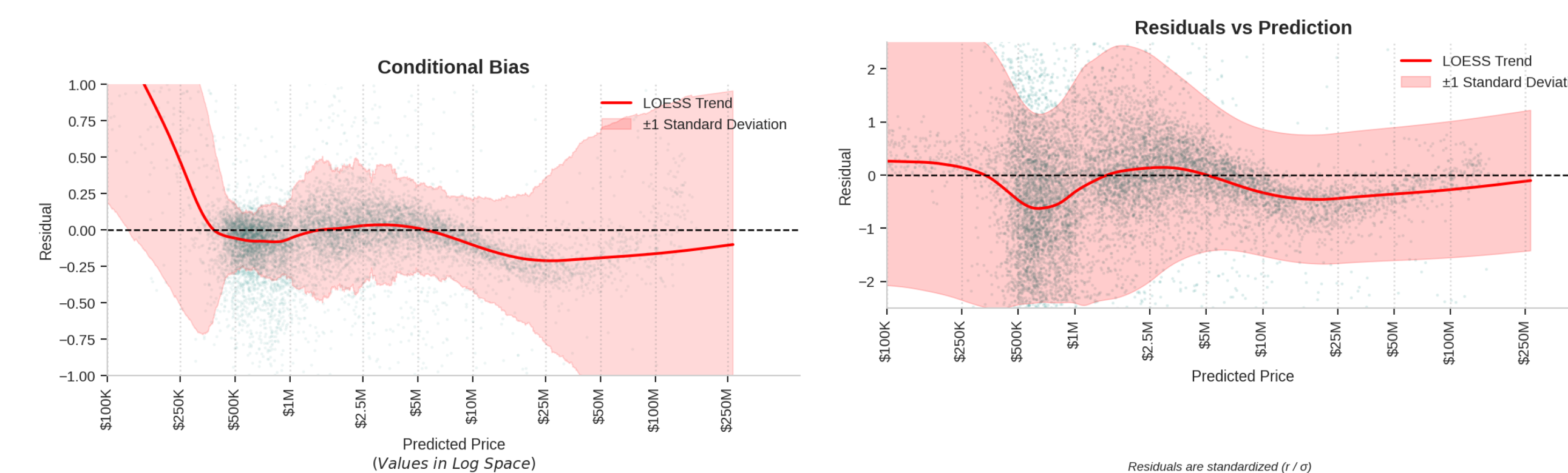
Table: Predictive metrics.

Residual Analysis



Kurtosis = 9.27 (Heavy Tails)

Figure: Diagnostics: Spatial clustering (left) and QQ Plot (right).



Bias vs. Prediction

Heteroscedasticity

Figure: Model Assumption Checks: Conditional Bias (left) and Homoscedasticity (right).

Latent Space Analysis

Primary Dimension (49.9% var): Encodes market trends and volume.

Secondary Dimension (30.2% var): Separates building classes.

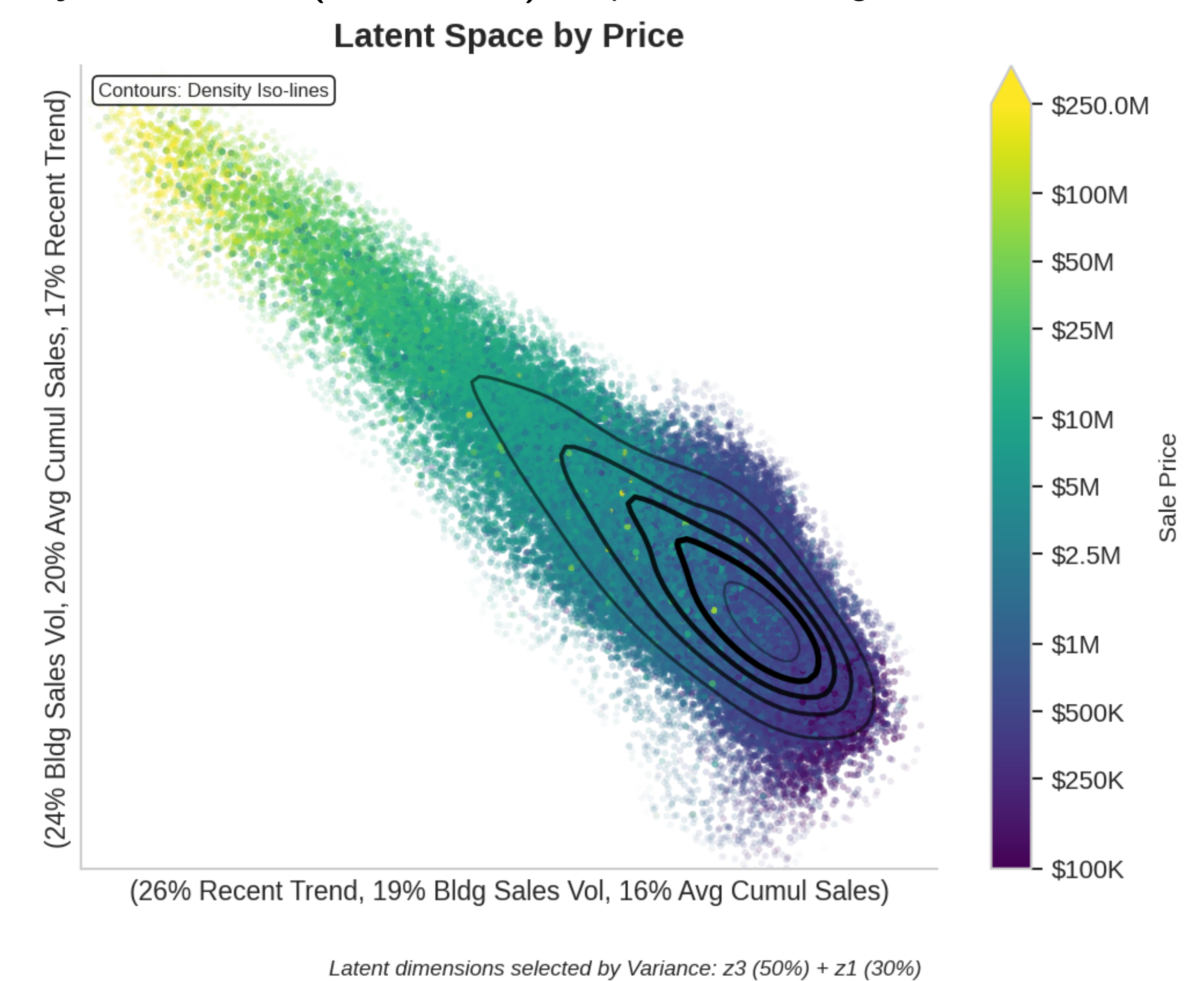


Figure: Latents by Price Decile: Smooth gradient tracks value.

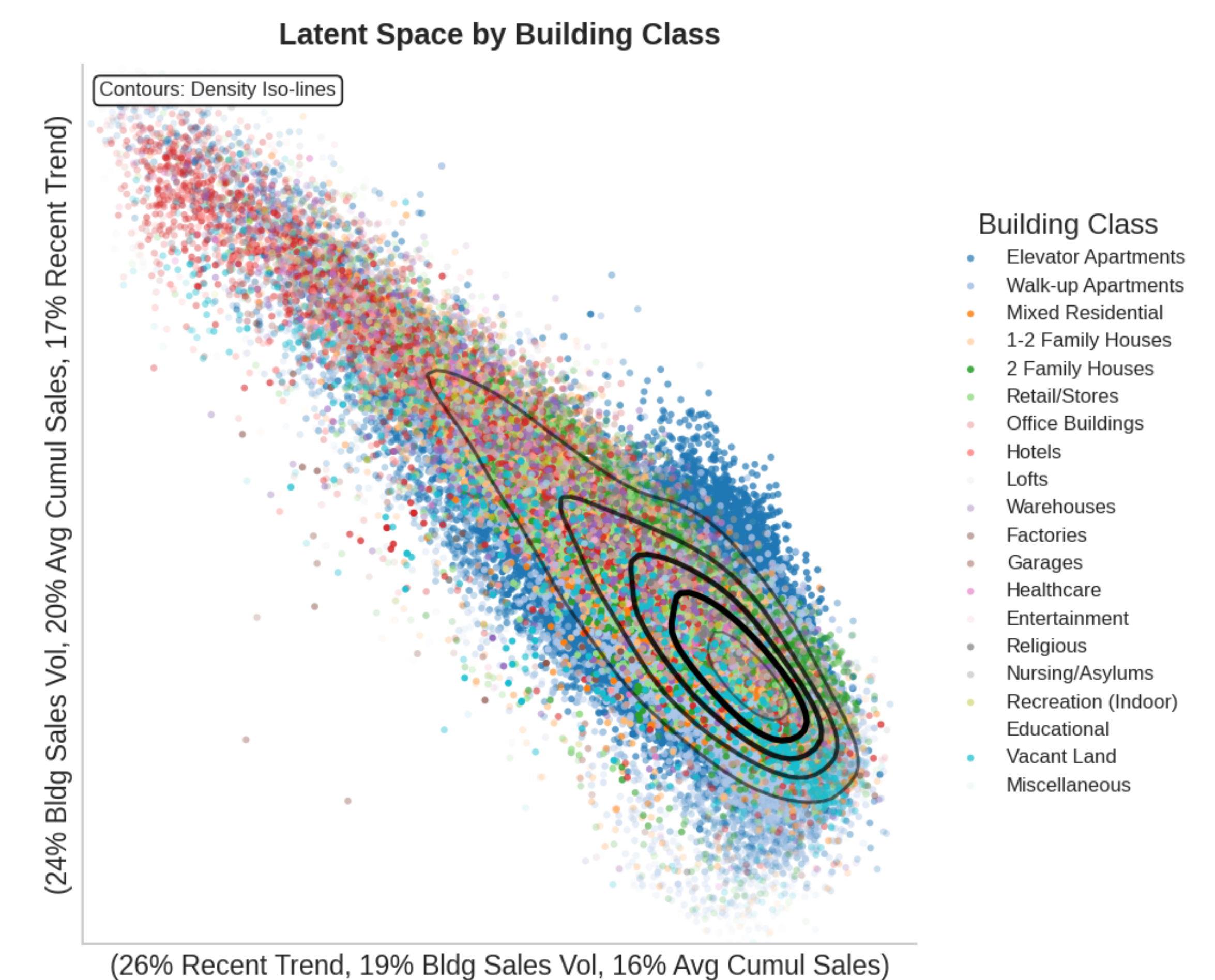


Figure: Latents by Building Class: Identifiable striations.